Semidefinite programming hierarchies for quantum adversaries

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(arXiv:1506.08810 - Quantum Bilinear Optimisation)







October 1st, 2015
QCrypt Tokyo

Overview

- Theoretical talk, plus start with non-cryptographic problem
- Classical noisy channel coding versus entanglement-assisted *channel coding* (**quantum assistance**)
- Semidefinite programming (sdp) hierarchies for understanding (bounding) the difference
- Cryptography: randomness extractors versus *quantum-proof randomness extractors* (**quantum adversary**)
- Conclusion / Outlook

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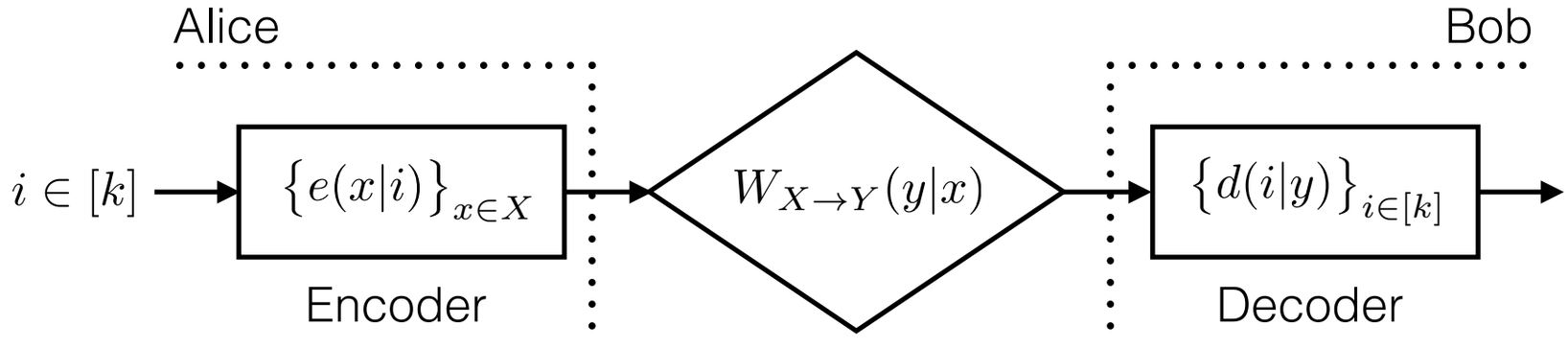
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Classical noisy channel coding (I)



- Given noisy channel $W_{X \to Y}$ mapping X to Y with transition probability:

$$W_{X \to Y}(y|x) \; \forall (x, y) \in X \times Y$$

- The goal is to send k different messages using W while minimising the error probability for decoding:

$$p_{\text{succ}}(W, k) := \underset{(e,d)}{\text{maximize}} \quad \frac{1}{k} \sum_{x,y,i} W_{X \to Y}(y|x) e(x|i) d(i|y)$$

"bilinear optimisation"

$$\text{subject to} \quad \sum_{x} e(x|i) = 1 \quad \forall i \in [k] \,, \quad \sum_{i} d(i|y) = 1 \quad \forall y \in Y$$

$$0 \leq e(x|i) \leq 1 \quad \forall (x, i) \in X \times [k] \,, \quad 0 \leq d(i|y) \leq 1 \quad \forall (i, y) \in [k] \times Y \,.$$

Classical noisy channel coding (II)

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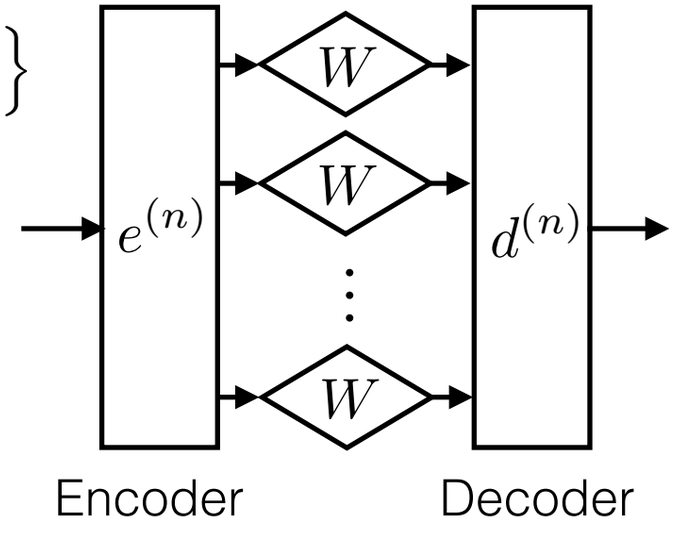
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compared to

- Shannon's asymptotic independent and identical distributed (iid) channel capacity:

Definition: $C(W) := \sup\left\{R \,\middle|\, \forall \delta > 0 : \lim_{n\to\infty} p_{\mathrm{succ}}\left(W^{\times n}, [R(1-\delta)]^n\right) = 1\right\}$

Answer: $C(W) = \max_{P_X} I(X:Y)$ mutual information



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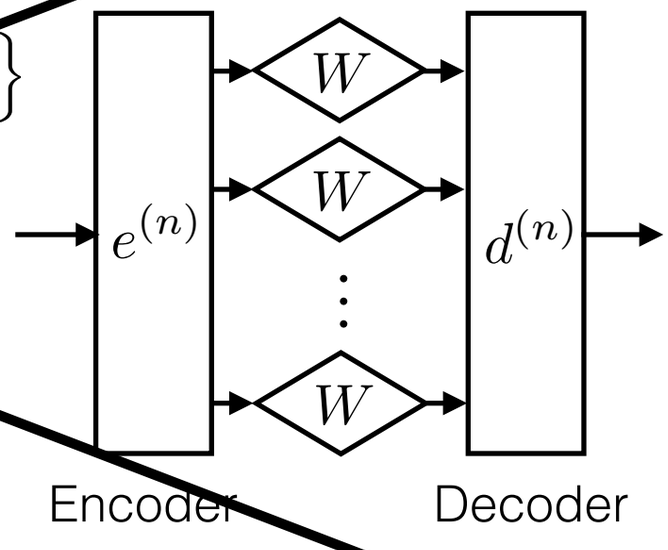
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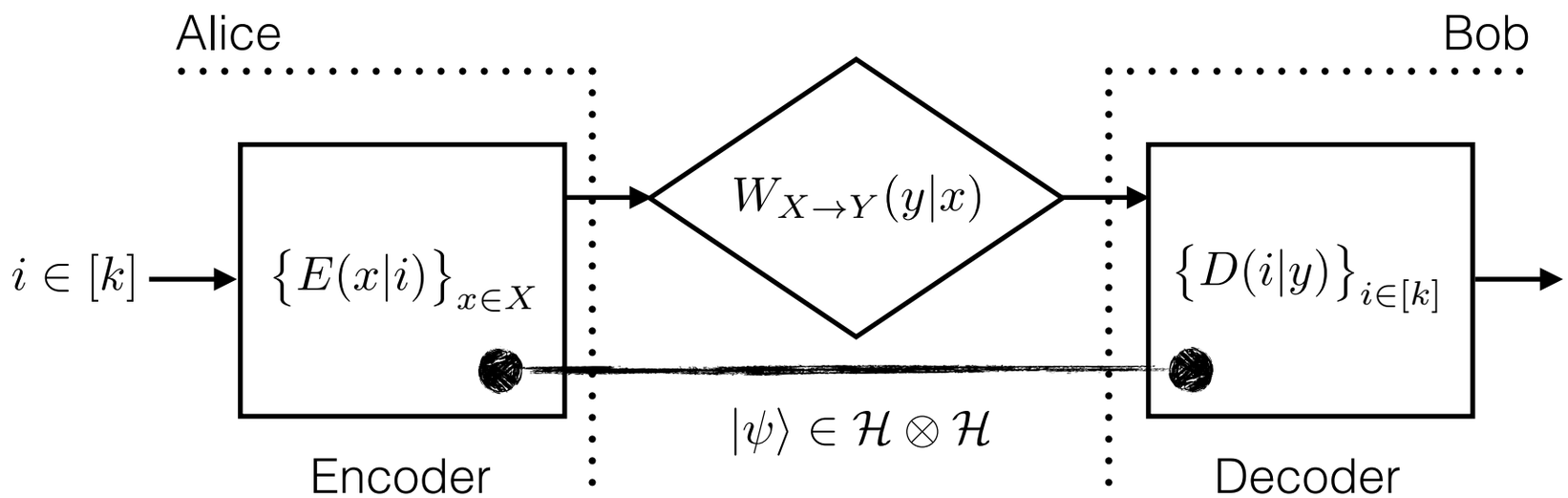
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Entanglement-assisted channel coding (I)



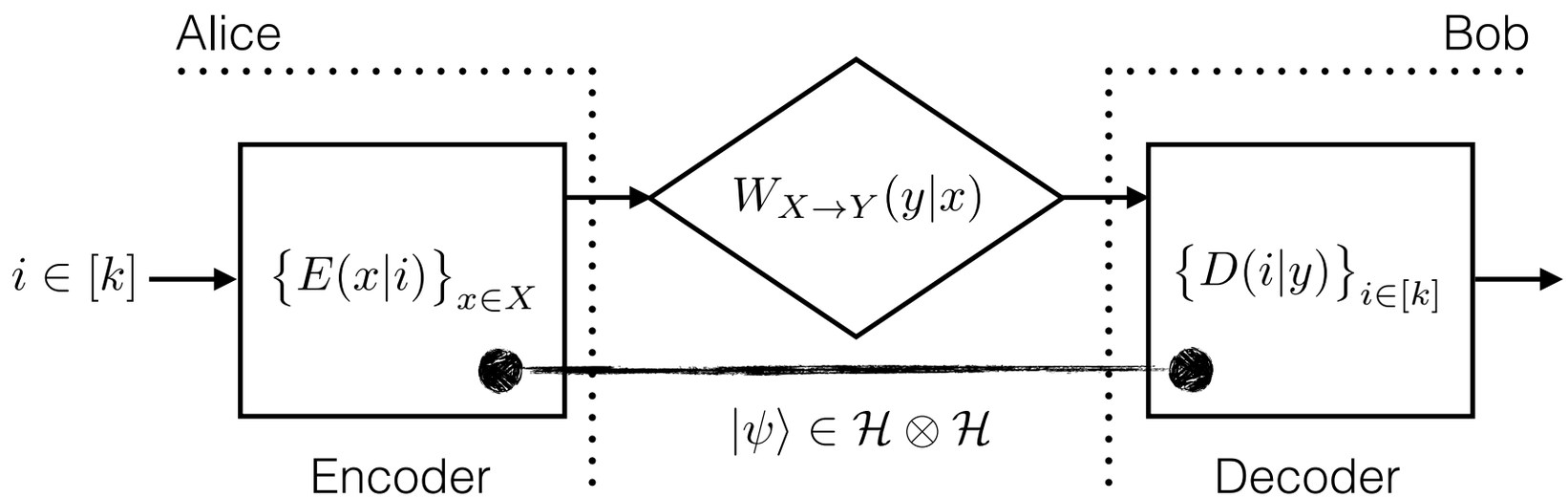
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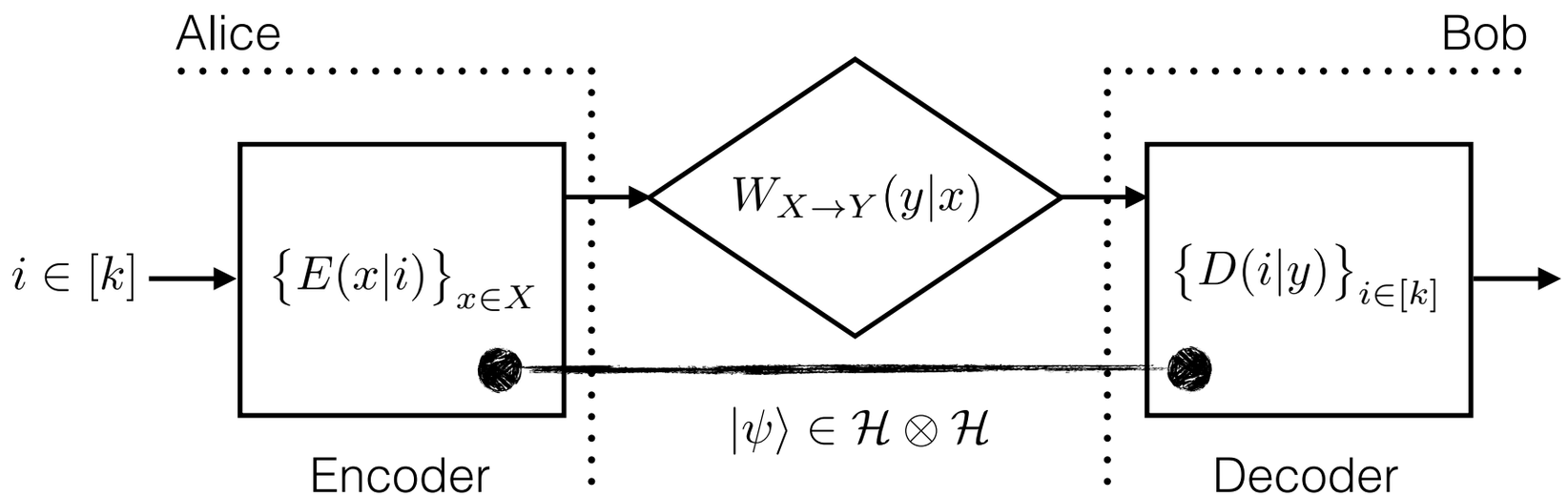
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- Unknown if $p^*_{\text{succ}}(W,k)$ is computable!

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- Understand the possible separation: $p_{\mathrm{succ}}(W,k) \quad \text{versus} \quad p^*_{\mathrm{succ}}(W,k)$

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$$p_{\mathrm{succ}}(Z,2) = \frac{5}{6} \approx 0.833 \quad \text{vs.} \quad p^*_{\mathrm{succ}}(Z,2) \geq \frac{2+2^{-1/2}}{3} \approx 0.902$$

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"linear optimisation with semidefinite constraints"

- We give a **converging hierarchy of semidefinite programming (sdp) relaxations**:

$$p_{\mathrm{succ}}(W,k) \leq p^*_{\mathrm{succ}}(W,k) = \mathrm{sdp}_\infty(W,k) \leq \ldots \leq \mathrm{sdp}_1(W,k) \quad \text{<— } \textit{efficiently computable!}$$

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- Quantum bilinear program:

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*motivated by: "**NPA hierarchy**" (Bell inequalities)*

[Lasserre, SIAM (2001)], [Parrilo, Math. Program. (2003)], [Navascues et al., PRL (2007)], [Doherty et al., IEEE CCC (2008)], [Navascues et al., NJP (2008)], [Pironio et al., SIAM (2010)]

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- First step: see as the part of the upper-right block of the Gram matrix

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for i=j

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- Original constraints can be formulated as positivity conditions on Ω : $\text{sdp}_1(W,k)$

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\text{subject to} \quad & \Omega \in \mathrm{Pos}(1+k|X|+k|Y|)\,, \quad \Omega_{\emptyset,\emptyset} = 1 \quad \text{with } \emptyset \text{ the empty symbol} \\
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- Going back to our example:

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p_{\mathrm{succ}}(Z,2) = \frac{5}{6} \approx 0.833
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(known before, with two-dimensional assistance)

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\mathrm{sdp}_1(W,k) = \underset{\Omega}{\text{maximize}} \quad & \frac{1}{k}\sum_{x,y,i} W_{X\to Y}(y|x)\Omega_{(i,x),(i,y)} \\
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- Going back to our example:

$$
Z = \begin{pmatrix}
1/3 & 1/3 & 0 & 0 \\
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$$
p_{\mathrm{succ}}(Z,2) = \frac{5}{6} \approx 0.833
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(known before, with two-dimensional assistance)

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First level semidefinite programming relaxation (II)

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—> further work [Barman and Fawzi., arXiv:1508.04095]

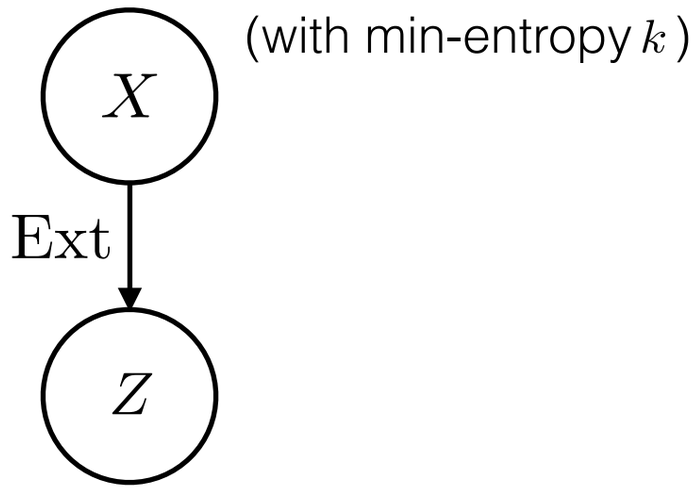
Overview

- Theoretical talk, plus start with non-cryptographic problem
- Classical noisy channel coding versus entanglement-assisted *channel coding* (**quantum assistance**)
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Quantum Cryptography (I)

- Privacy amplification:

  weak source of randomness $X \in \{0,1\}^n$



  uniform random bits $Z \in \{0,1\}^m$

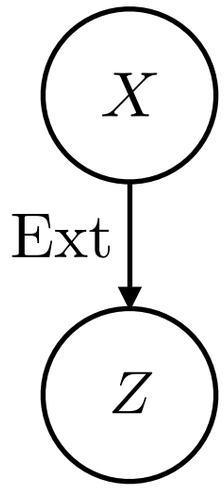
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- Example: two-universal hashing

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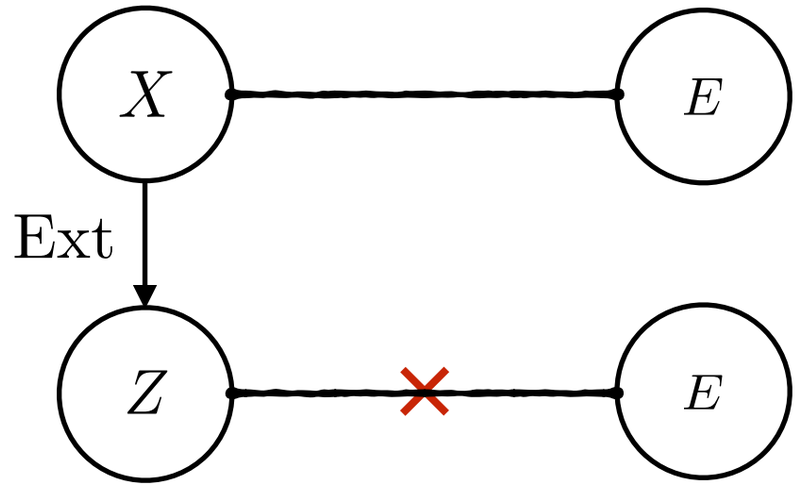
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- What happens for quantum adversaries?

  weak source of randomness relative to E



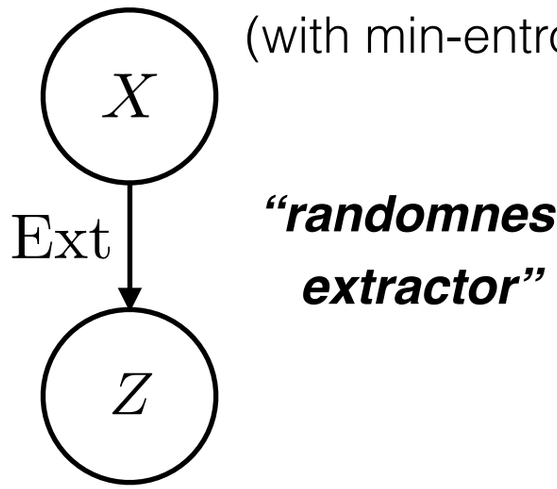
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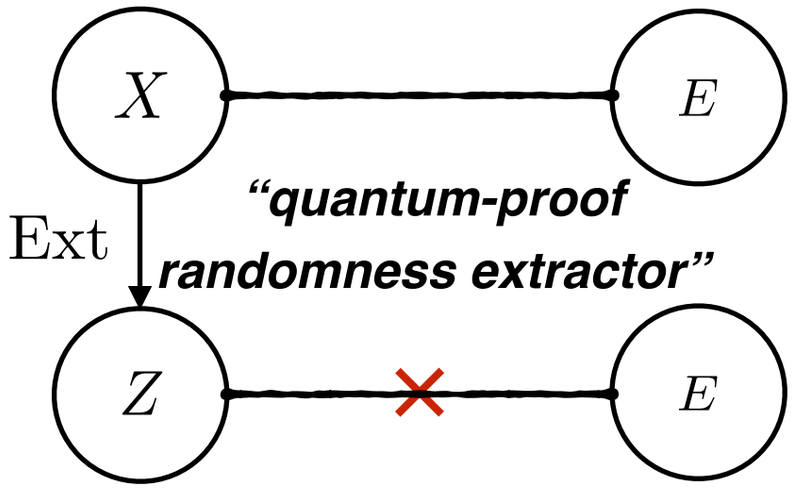
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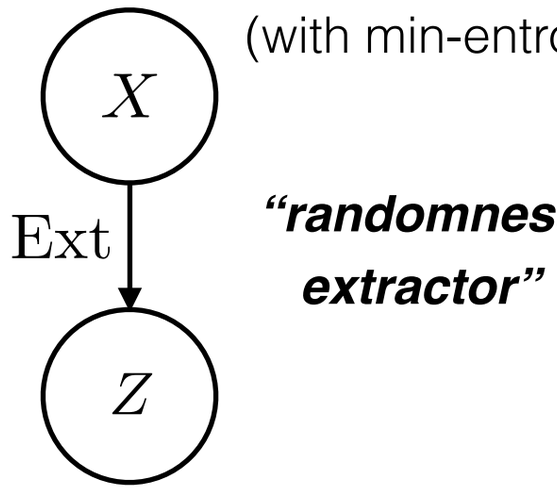
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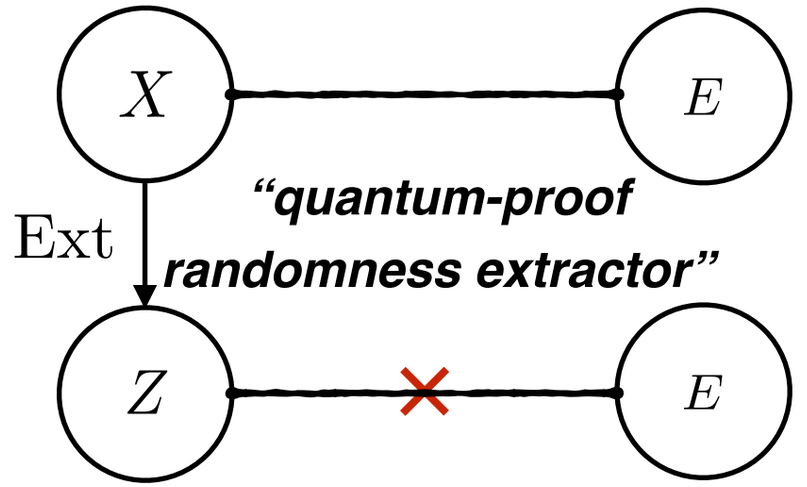


"randomness extractor"

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"quantum-proof randomness extractor"

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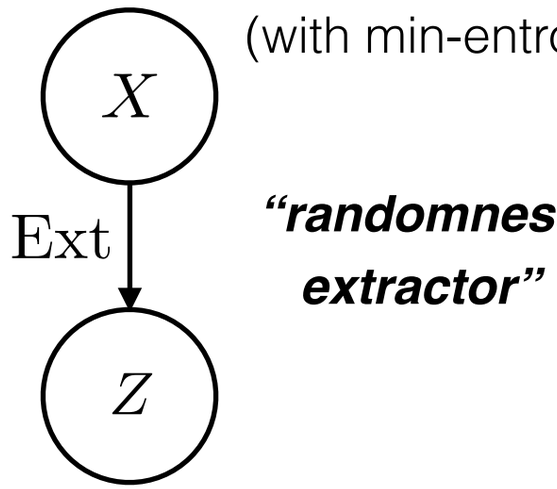
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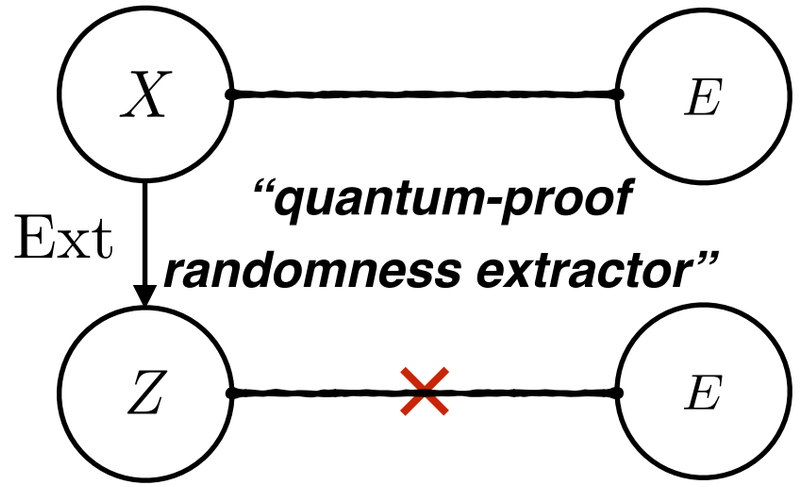


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- Classical versus quantum error (the ϵ): $C(\mathrm{Ext}, k)$ versus $Q(\mathrm{Ext}, k)$ *computable?*

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- For randomness extractors: classical versus **quantum adversaries**

$C(\mathrm{Ext}, k)$ versus $Q(\mathrm{Ext}, k)$ *computable?*

classical bilinear optimisation	versus	quantum bilinear optimisation
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- Converging **hierarchy of semidefinite programming** relaxations:

$$C(\mathrm{Ext}, k) \leq Q(\mathrm{Ext}, k) = \mathrm{sdp}_\infty(\mathrm{Ext}, k) \leq \ldots \leq \mathrm{sdp}_1(\mathrm{Ext}, k)$$ <— *efficiently computable!*

—> upper bounding the power of quantum adversaries

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- See our paper for results: arXiv:1506.08810 - Quantum Bilinear Optimisation

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- Two-prover games (**Bell inequalities**): we get tighter hierarchy (than previous work)

  —> first level also in independent work [Sikora and Varvitsiotis, arXiv:1506.07297]

- Optimisations over the **completely positive semidefinite cone**: we get the first hierarchy (quantum graph parameters)

  [Laurent and Piovesan, arXiv:1312.6643]

Thanks!